Release Notes for the Highway-Runoff Database (HRDB) Version 1.1.0, a data warehouse and preprocessor for the Stochastic Empirical Loading and Dilution Model

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Summary

The Highway-Runoff Database (HRDB) was developed by the U.S. Geological Survey (USGS), in cooperation with the Federal Highway Administration (FHWA) Office of Project Delivery and Environmental Review to provide planning-level information for decision makers, planners, and highway engineers to assess and mitigate possible adverse effects of highway runoff on the Nation's receiving waters (Granato and Cazenas, 2009; Granato, 2013; Granato and others, 2018). The HRDB was assembled by using a Microsoft Access database application to facilitate use of the data and to calculate runoff-quality statistics with methods that properly handle censored-concentration data. The HRDB was first published as version 1.0 in cooperation with the FHWA in 2009 (Granato and Cazenas, 2009). The second version (1.0.0a) was published in cooperation with the Massachusetts Department of Transportation (DOT) Highway Division to include data from Ohio and Massachusetts (Smith and Granato, 2010). The third version (1.0.0b) was published in cooperation with the FHWA to include a substantial amount of additional data (Granato and others, 2018; Granato and Jones, 2019). This version includes data from 242 highway sites across the country; data from 6,837 storm events; and 106,441 concentration values with data for 414 different water-quality constituents. This new version (1.1.0) of the database contains software updates to provide data-quality information within the Graphical User Interface (GUI), calculate statistics for multiple sites in batch mode, and output additional statistics.

Purpose

The HRDB was developed to serve as a data warehouse for current and future highway-runoff data sets (Granato and Cazenas, 2009; Granato, 2013; Granato and others, 2018). The database can be used to document information about a data set, monitoring site(s), and highway-runoff data (including precipitation, runoff, and event-mean concentrations of water-quality constituents). The HRDB provides information and data that may be used to assess potential effects of highway runoff on receiving waters and the need for management measures to mitigate the potential for adverse effects on receiving waters. The HRDB application also was developed to serve as a data preprocessor for the Stochastic Empirical Loading and Dilution Model (SELDM).

New Forms in the Graphical User Interface

This new version (1.1.0) is a software update that includes minor changes to the Visual Basic for Applications (VBA) code that runs the database application's graphical user interface (GUI). This is a software upgrade only; no new data were added to the previous version 1.0.0b (Granato and others, 2018). Use of the HRDB GUI is described in detail by Granato and Cazenas (2009). A new form, the Data-Set-Quality Documentation Form, was added to provide data-quality summaries within the GUI (figs. 1 and 2). A new form also was added to allow the user to calculate concentration statistics for multiple sites by grouping the data or, in the new version, by calculating statistics for each site in the selected group separately (figs. 1 and 3). The option to show or hide alternative statistics for concentrations with censored values was added as a checkbox to the plotting position formula selection form, which appears before the water-quality statistics output form (fig 1).

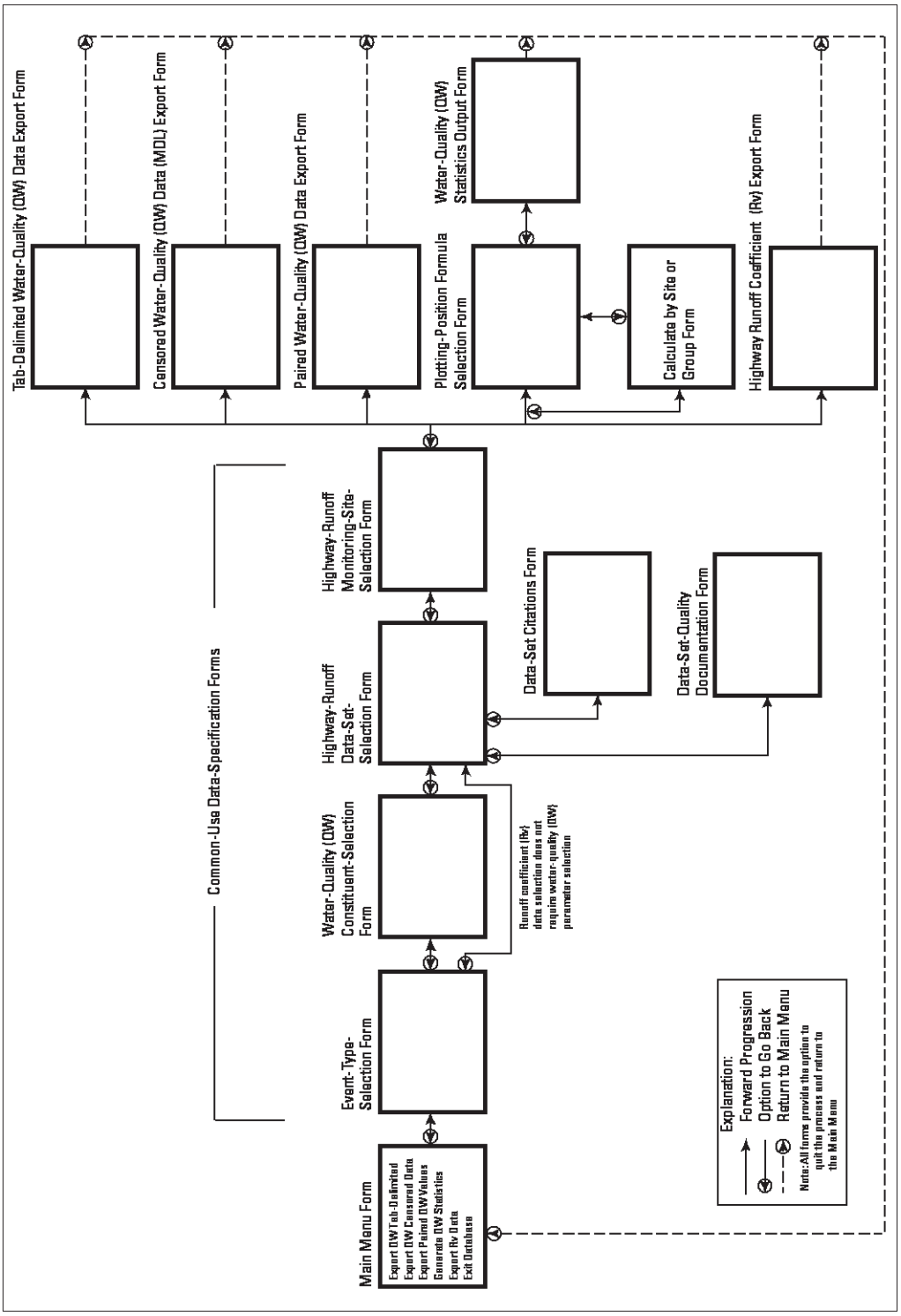


Figure 1. Highway-runoff database application schematic for processing water-quality (QW) and runoff-coefficient (Rv) data (modified from Granato and Cazenas, 2009).

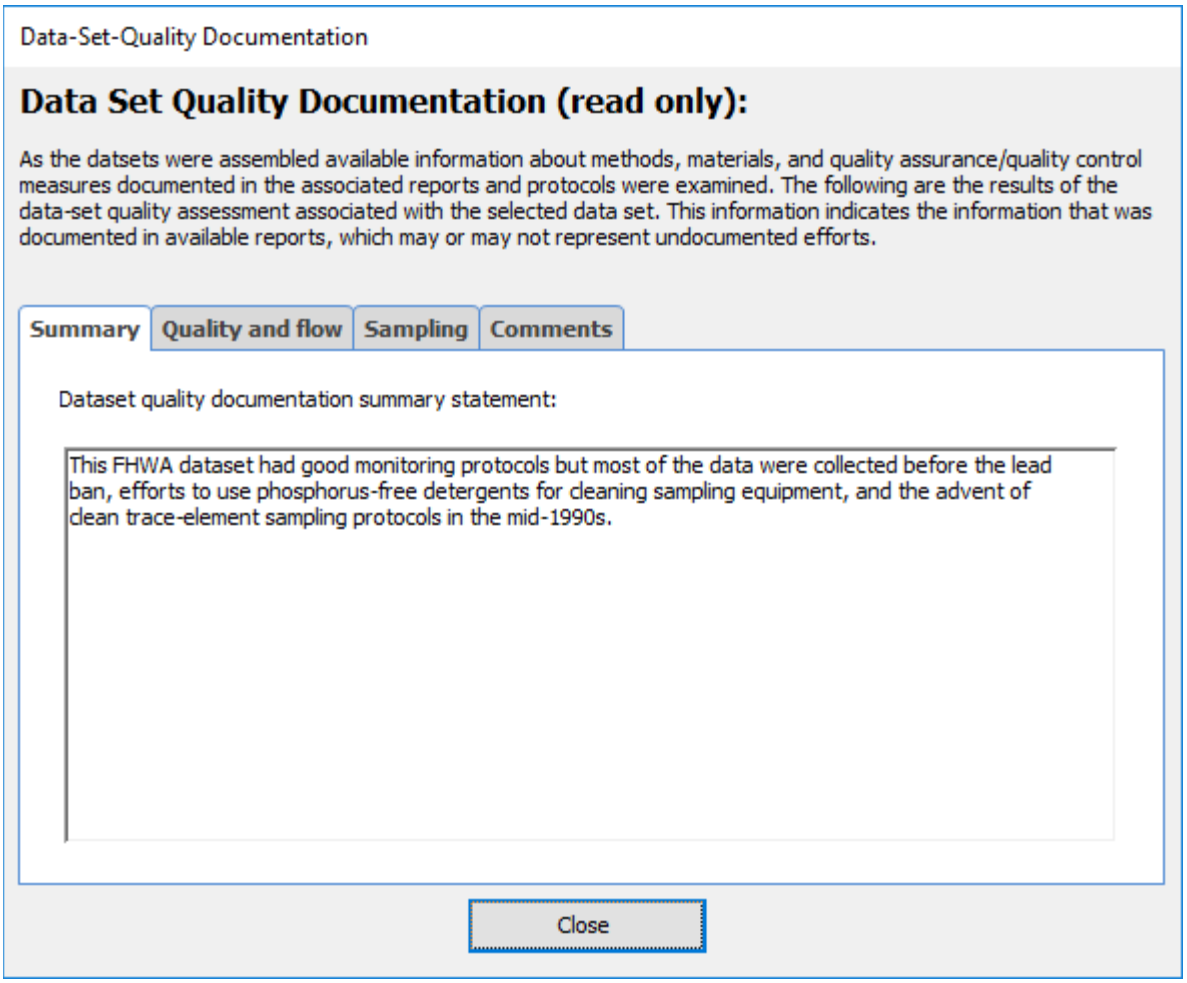


Figure 2. Data Set Quality Documentation form showing the summary information for the 1990 Federal Highway Administration data set.

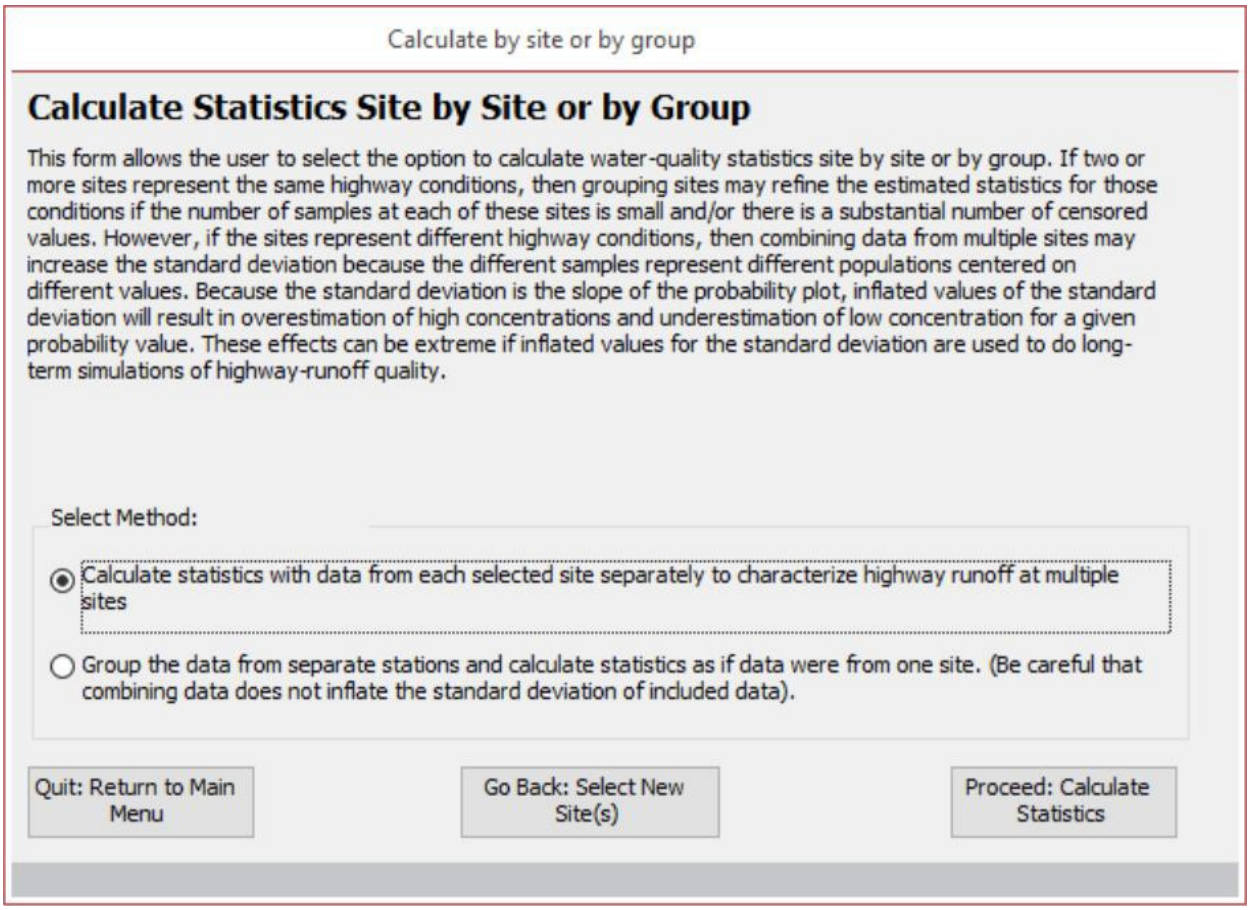


Figure 3. Calculate Statistics by Site or by Group form showing the selection of the by-site option.

The data and information in the HRDB were compiled from multiple sources including reports and data from the FHWA, USGS, State agencies, universities and colleges, and individual researchers (Granato and others, 2018). The processes for incorporating these data were subjected to rigorous review and the information in the database is a substantially complete and accurate representation of the contents of the source materials. Available information, however, was not sufficient to verify all datasets. It is difficult to collect representative runoff-quality data (Granato and others, 2003; by Granato and Cazenas, 2009; Smith and Granato, 2010). Therefore, data were provided without warranty, expressed or implied, as to accuracy or completeness

(Granato and others, 2018). Data-quality metadata were recorded for each dataset, but these data were not provided within the GUI. The Data-Set-Quality Documentation Form (fig. 2), which is accessed from the Highway-Runoff Data-Set-Selection Form (fig 1.), provides a data-quality summary and information about runoff quality and flow measurement, sampling methods, and other data-quality comments.

The option to calculate statistics by site or by group was added to facilitate calculation of statistics from many sites without having to cycle back and forth between the water-quality statistics output form and either the site selection or the data-set selection form. Use of the HRDB in interpretive studies revealed the need for efficiently calculating multiple statistics for many sites (Taylor and others, 2014; Granato and Jones, 2015, 2017, 2019; Weaver and others, 2019). The users could export the data to calculate statistics independently, but the HRDB can calculate statistics for datasets with censored values by using the Regression on Order Statistics (ROS) method (Granato and Cazenas, 2009). Both the by-site and by-group options are available because if two or more sites represent the same highway conditions, then grouping sites may refine the estimated statistics for those conditions if the number of samples at each site is small and/or there is a substantial number of censored values. Conversely, if the sites represent different highway conditions, then combining data from multiple sites may increase the standard deviation because each site represents independent populations centered on different values. Because the standard deviation is the slope of the probability plot, inflated values of the standard deviation will result in overestimation of high concentrations and underestimation of low concentration for a given probability value. These effects can be extreme if inflated values for the standard deviation are used to do long-term simulations of highway-runoff quality (Granato, 2013; Smith and others, 2018).

New Water-Quality Statistics

Six new statistics were added to the water-quality algorithms. These statistics are Pearson's second skew and the standard errors of the average, median, standard deviation, skew, and the standard error of the Pearson's second skew. These statistics were added to the HRDB by using equations from the literature (Haan, 1977; U.S. Interagency Advisory Committee on Water Data, 1982; Doane and Seward, 2011; Harding and others, 2014). Most water-quality values have a lower bound of zero and vary by orders of magnitude in highway and urban runoff so the HRDB calculates statistics in arithmetic and logarithmic space (Granato and Cazenas, 2009; Granato, 2013; Granato and Jones, 2015, 2017, 2019; Smith and others, 2018).

Pearson's second skew, which is calculated by subtracting the median from the average, multiplying the difference by 3, and dividing by the standard deviation was added because this measure of skew is more robust to extreme values than the traditional (Fisher–Pearson) skew also used in the HRDB (Haan, 1977; Doane and Seward, 2011; Smith and others, 2018). The ROS method used in the HRDB is designed to estimate the average and standard deviation of left censored datasets based on a lognormal distribution, but it may not represent the skew of estimated values (Granato and Cazenas, 2009). Because the traditional equation for skewness uses the sum of cubed differences from the mean and the cubed standard deviation (Haan, 1977; U.S. Interagency Advisory Committee on Water Data, 1982), estimation of low outliers may affect the traditional skew.

The standard error of each of these statistics represents the variability that would occur if repeated samples were drawn from the underlying population (Haan, 1977; U.S. Interagency Advisory Committee on Water Data, 1982; Harding and others, 2014). Standard errors can be used to calculate confidence intervals for each statistic. If the confidence intervals of statistics from

different sites do not overlap, then values are interpreted as statistically different at the given confidence interval (Gardner and Altman, 1986; Goodman, 2008; Lesaffre, 2008). If the confidence interval of the skew statistics includes values that are greater than and less than or equal to zero, then the datasets can be interpreted to be normal if the skew of the arithmetic values meet these criteria, or lognormal if skew of the log-transformed values meet these criteria (Haan, 1977; Granato, 2013; Smith and others, 2018). If the confidence interval of the traditional skew and Pearson's second skew overlap, then they are not statistically different. Standard errors are not estimated for the statistics calculated by using simple substitution of censored values because substitution of a standard value has adverse effects on the standard-error estimators. The standard error equations are:

Standard error of the average:

SEA = SD/N^0.5

Standard error of the median:

SEMD = (SD/N^0.5)*(PI/2)^0.5

Standard Error of the standard deviation:

SESD = SD * (1+0.75*Skew^2)^0.5 / (2 * N)^0.5

Standard Error of the skew:

SESkew = ((6*N*(N-1))/((N-2)*(N+1)*(N+3)))^0.5

Standard Error of Pearson's second skew:

SEP2skew = (3/N^0.5)*((PI/2)-1)^0.5

Where

SD is the standard deviation;

N is the number of values in the sample;

Skew is the traditional (Fisher–Pearson) skew; and

PI is the mathematical constant Pi rounded to double precision for VBA (3.14159265358979) in the HRDB.

To obtain the confidence intervals for these statistics the standard error is multiplied by the student's t-value for the desired confidence level and this value is added or subtracted from the statistic of interest. The HRDB calculates the standard error rather than the confidence limits because several different confidence limits commonly are used in hydrological investigations. The student's t-value for the 95-percent confidence limit ranges from 12.706 with two samples to the commonly used 1.960 at about 4,500 samples. About 39 percent of site-samples (a given constituent for a given site) in the HRDB include less than 10 values (with a corresponding t of 2.262); 50 percent include less than 12 samples (with a corresponding t of 2.201); 84 percent include less than 20 samples (with a corresponding t of 2.093); and 99 percent have less than 62 values (with a corresponding t of about 2.000). Both the standard errors and the student's t-values decrease with increasing sample size, which has a multiplicative effect decreasing the range of the confidence intervals.

32-Bit Microsoft Access vs 64-Bit Microsoft Access

This application is developed for the 32-bit version of Microsoft Access. Microsoft recommends the 32-bit version of Office for most users, because it is compatible with most other applications (Microsoft Corporation, 2016, 2018; 404 Tech Support Inc, 2017). Hence, the 32-bit version of Office is the default installation for 32- and 64-bit versions of Windows. The 32-

bit Office client is supported as a Windows-32-on-Windows-64 (WOW64) installation. WOW64 is the x86 emulator that enables 32-bit Windows-based applications to run seamlessly on 64-bit Windows systems. This allows users to continue using existing Microsoft ActiveX Controls and COM add-ins with 32-bit Office. Only users who must manipulate extremely large recordsets (such as spreadsheets with more than one-million rows) would need to use the 64-bit version of Office. Furthermore, 64-bit files are not backward compatible. Therefore, this database application was created and compiled by using the 32-bit version of MS Access. The installed runtime version of the 32-bit application should run, but the compiled version will not run if the user has the 64-bit version of Office installed.

If you have the 64-bit version of MS Office use the decompiled version of the 32-bit database (.accdb), copy it over to a 64-bit version of MS Access. Compact and repair the database, then compile the code. At this point you will need to address any errors (such as missing references) and once you have successfully compiled the database, compact and repair it again. You may make an executable version but keep an uncompiled version for future updates.

Rights

The Highway-Runoff Database (HRDB) was prepared by the U.S. Geological Survey (USGS) in cooperation with the Federal Highway Administration (FHWA), which are agencies of the U.S. Federal government. The database was compiled from multiple sources including reports and data from the FHWA, State agencies, universities and colleges, and individual researchers. Although the information, metadata, and computer files have been subjected to rigorous review and are substantially complete and accurate, this information is presented without warranty, expressed or implied, as to accuracy or completeness. The USGS and the

FHWA reserve the right to revise this database pursuant to further analysis and review. This database has been used by the USGS and is released on condition that the USGS or the FHWA may not be held liable for any damages resulting from their use. Information and numerical methods used in this relational database are attributable to the cited works. Copyright protection under Title 17 is not available for any work of the U.S. Government (17 USC §105). This database application is provided solely for the purposes of private study, scholarship, or research. Therefore, the information in this database is in the public domain for these purposes but may not be available for some commercial uses. By using the HRDB, you agree to appropriately acknowledge the authors, the USGS, and FHWA in publications that result from the use of this database or in products that include this database in whole or in part. Because the database and related material are free and provided "as is," the authors, the USGS, the FHWA, and the United States Government are not obligated to provide users with any support, consulting, training, or assistance of any kind with regard to the use, operation, and performance of this database nor to provide users with any updates, revisions, new versions, or "bug fixes". The use of trade or product names in this database is for identification purposes only and does not constitute endorsement by the U.S. Government.

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